

Congruence, Similarity and Geometrical Proof



DR ESLAM AHMED
eliteigcse.com

Similar shapes: same angles, sides in one ratio — find the scale factor and use it everywhere. Congruent: exactly the same size. Prove it with one of four conditions.

THE TOPIC ON ONE MAP

Congruence, Similarity and Geometrical Proof

01 — Similar shapes: the scale factor

new side = old $\times k$

match equal angles

02 — Similar triangles in a diagram

parallel lines

an X shape

nested triangles

03 — Proof

SSS, SAS, ASA, RHS

prove similar: two angles

TWO WORDS Know the difference:

ZONE SIMILAR

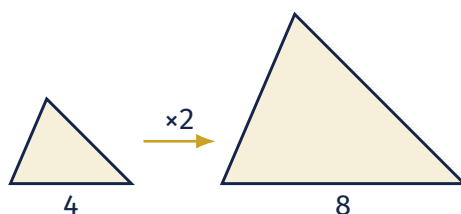
same shape

equal angles; sides in the same ratio k

ZONE CONGRUENT

same shape and size

SSS, SAS, ASA (AAS), RHS



- 1 match the equal angles
- 2 $k = \frac{\text{new}}{\text{old}}$ from one pair
- 3 big = small $\times k$, small = big $\div k$

01 Similar Shapes: the Scale Factor

Pair the sides that sit opposite equal angles. One complete pair gives the scale factor k ; every other side uses the same k .

WHEN YOU SEE → YOU DO

“mathematically similar”

$$\rightarrow k = \frac{\text{new length}}{\text{old length}}$$

the shapes are turned or flipped

→ match by the angles, not by position

a scale model

$$\rightarrow k = \frac{\text{model}}{\text{real}}$$

“write y in terms of x ”

$$\rightarrow y = kx$$

WORKED MODEL • NOVEMBER 2021 P2H

$ABC \sim DEF$: $AC = 45$, $DF = 20$, $DE = 36$. Find AB .

1 $k = \frac{45}{20} = 2.25$ (small → large)

matching pair

ANSWER $AB = 36 \times 2.25 = 81 \text{ cm}$

× **TRAP** Pairing sides by where they are drawn. Pair them by the equal angles.

IDEA 01

Practise: Similar Shapes: the Scale Factor

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Scale factor

CH 4 · TOPIC 8 2 marks

Two similar triangles have corresponding sides 6 cm and 15 cm. Another side of the small one is 4 cm. Find the matching side of the large one.

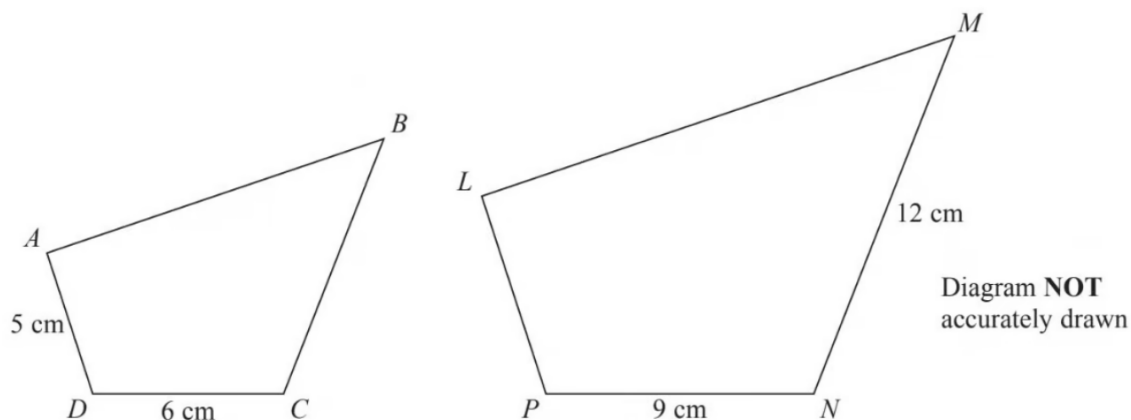
SELECTED PRACTICE

Q2**PRACTICE**

Practice · Easy

CH 4 · TOPIC 8

4 marks

1 (a)

Quadrilaterals $ABCD$ and $LMNP$ are mathematically similar.

Angle A = angle L

Angle B = angle M

Angle C = angle N

Angle D = angle P

Work out the length of LP .

(2 marks)

(b) Work out the length of BC .

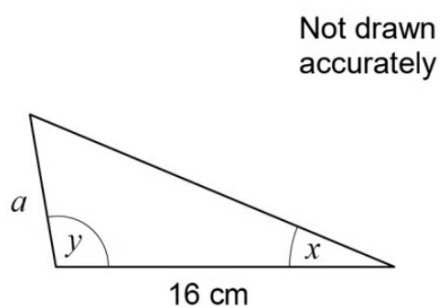
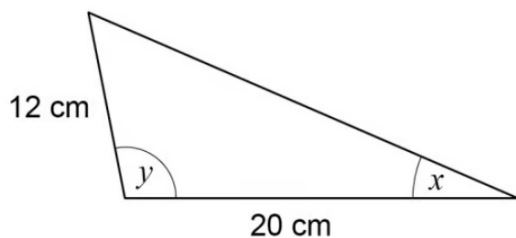
(2 marks)

Q3**PRACTICE**

Practice · Easy

CH 4 · TOPIC 8 **2 marks**

- 8 These two triangles are similar.



Work out the value of a .

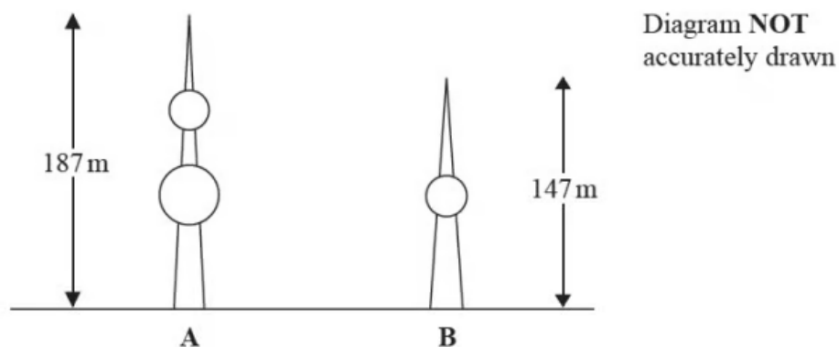
.....cm

(2 marks)

Q4 PRACTICE Practice · Medium

CH 4 · TOPIC 8 3 marks

- 3 The diagram shows two water towers in Kuwait.



The real height of tower **A** is 187m.

The real height of tower **B** is 147m.

Ahmed makes a scale model of both towers.

The height of tower **A** on the scale model is 90cm.

Work out the height of tower **B** on the scale model.

Give your answer correct to the nearest centimetre.

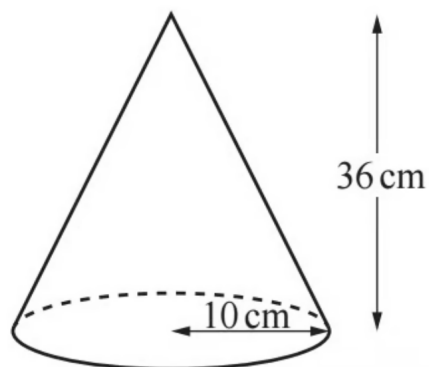
(3 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 8 6 marks

6 (a)

NOT TO
SCALE

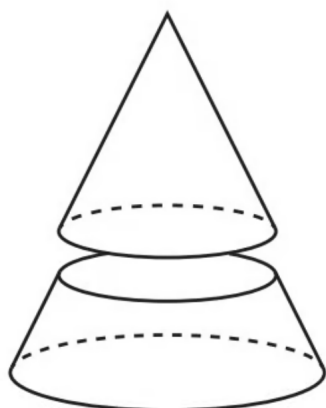
A solid metal cone has radius 10 cm and height 36 cm.
Calculate the volume of this cone.

[The volume, V , of a cone with radius r and height h is $V = \frac{1}{3} \pi r^2 h$.]

..... cm³

(2 marks)

(b) The cone is cut, parallel to its base, to give a smaller cone.

NOT TO
SCALE

The volume of the smaller cone is half the volume of the original cone.

The smaller cone is melted down to make two different spheres.

The ratio of the radii of these two spheres is 1 : 2.

Calculate the radius of the smaller sphere.

[The volume, V , of a sphere with radius r is $V = \frac{4}{3} \pi r^3$.]

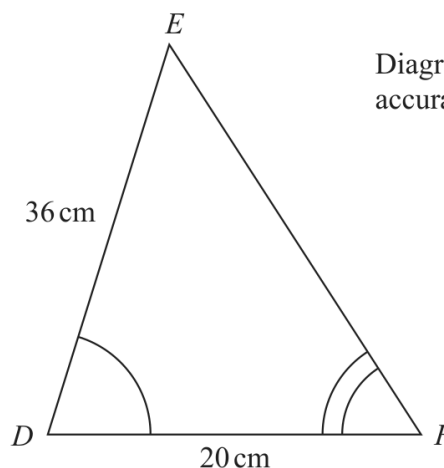
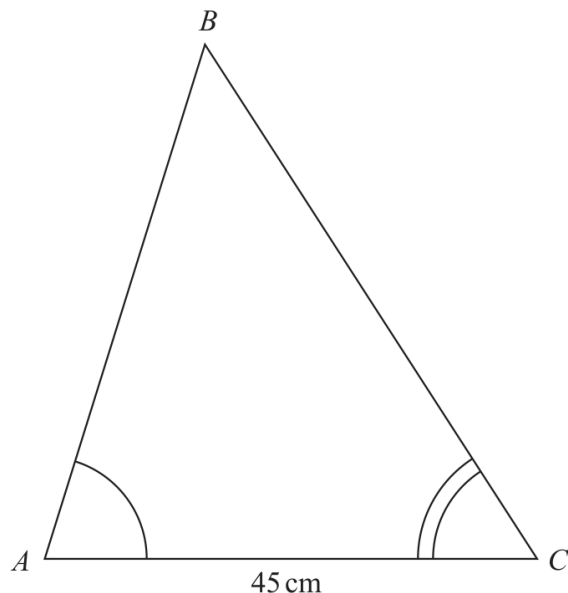
..... cm

(4 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 · November 2021 · Paper 2H · Q4

CH 4 · TOPIC 8 4 marks

4 ABC and DEF are similar triangles.Diagram **NOT**
accurately drawn(a) Work out the length of AB .

..... cm

Given that $BC = 54\text{ cm}$,(b) work out the length of EF .

..... cm

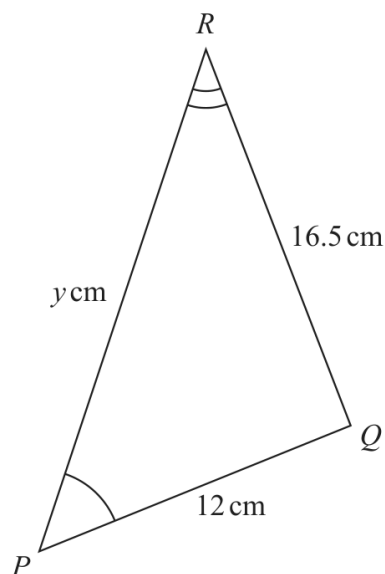
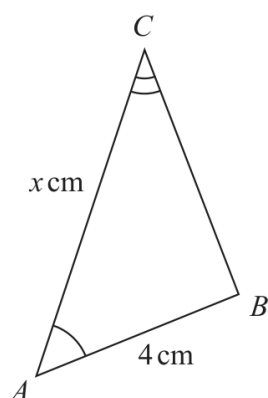
(Total for Question 4 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q3

CH 4 • TOPIC 8 3 marks

3

Diagram **NOT**
accurately drawnTriangle ABC is similar to triangle PQR $AB = 4\text{ cm}$ $PQ = 12\text{ cm}$ $RQ = 16.5\text{ cm}$ $AC = x\text{ cm}$ $PR = y\text{ cm}$ (a) Calculate the length of BC

..... cm

(b) Write down an expression for y in terms of x $y =$ **(Total for Question 3 is 3 marks)**

Q8 PAST PAPER Edexcel 4MA1 · May 2023 · Paper 1HR · Q3

CH 4 · TOPIC 8 4 marks

3 $ABCD$ and $EFGH$ are similar quadrilaterals.

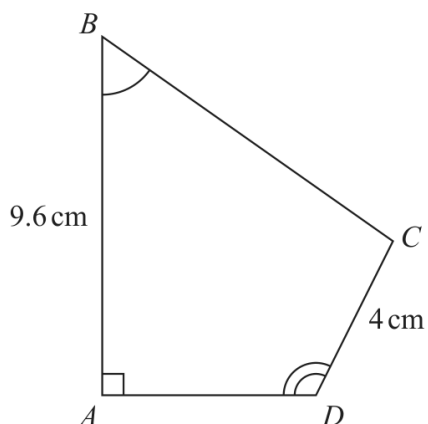
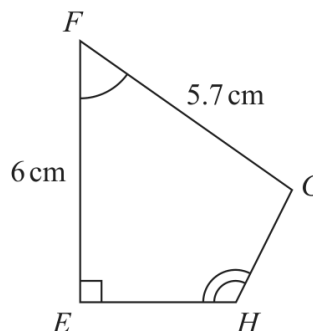


Diagram **NOT**
accurately drawn



(a) Work out the length of GH

..... cm

(b) Work out the length of BC

..... cm

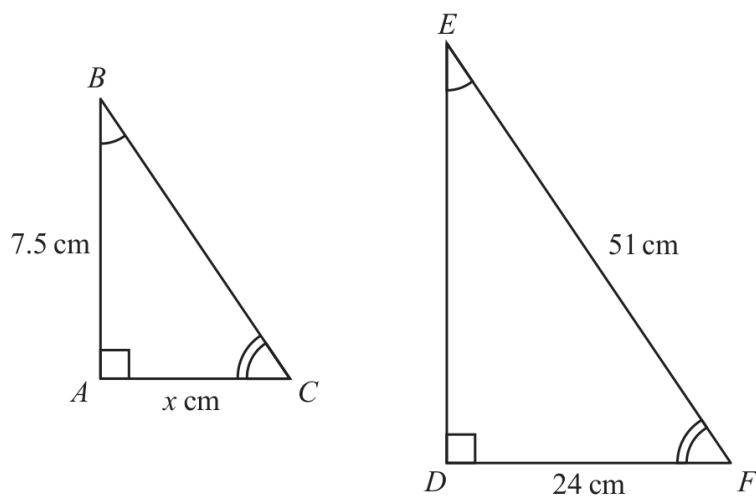
(Total for Question 3 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q10

CH 4 · TOPIC 8 5 marks

10 Here are two similar triangles.



Diagrams **NOT**
accurately drawn

Work out the value of x
Show your working clearly.

$x = \dots\dots\dots$

(Total for Question 10 is 5 marks)

Q10**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q5

CH 4 · TOPIC 8 4 marks

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

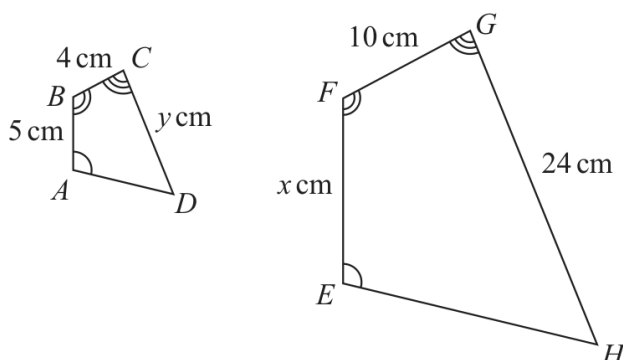


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm} \quad BC = 4 \text{ cm} \quad CD = y \text{ cm}$$

$$EF = x \text{ cm} \quad FG = 10 \text{ cm} \quad GH = 24 \text{ cm}$$

- (a) Work out the value of x

$$x = \dots\dots\dots$$

- (b) Work out the value of y

$$y = \dots\dots\dots$$

(Total for Question 5 is 4 marks)

Q11

PAST PAPER

Edexcel 4MA1 · November 2025 · Paper 2H · Q4

CH 4 · TOPIC 8 3 marks

4 Here are three similar quadrilaterals.

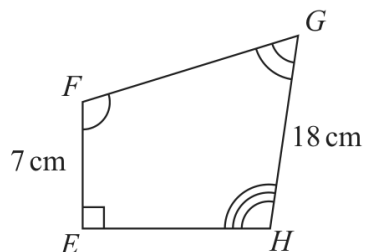
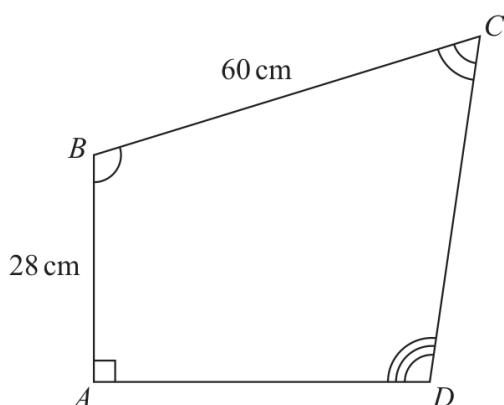
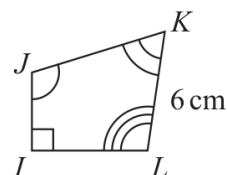


Diagram **NOT** accurately drawn



Work out the length of JK

..... cm

(Total for Question 4 is 3 marks)

02 Similar Triangles in a Diagram

Parallel lines make similar triangles: equal alternate or corresponding angles. Draw the two triangles separately, the same way round, then find k .

WHEN YOU SEE → YOU DO

$BE \parallel CD$ inside one triangle

→ nested: small ABE , large ACD

an X shape with $AB \parallel DE$

→ two triangles meeting at the crossing

“work out PB ” (a part of a side)

→ find the whole side first, then subtract

a right angle and a perpendicular

→ three similar triangles

WORKED MODEL · JUNE 2024 P2HR

$BE \parallel CD$; $AE = 10$; $CD = 1.5 \times BE$. Find ED .

1 $k = 1.5$, so $AD = 1.5 \times 10 = 15$

nested triangles

ANSWER $ED = 15 - 10 = 5 \text{ cm}$

IDEA 02

Practise: Similar Triangles in a Diagram

My questions first, then selected practice, then real past papers.

Q12 DR ESLAM Nested triangles

CH 4 · TOPIC 8 3 marks

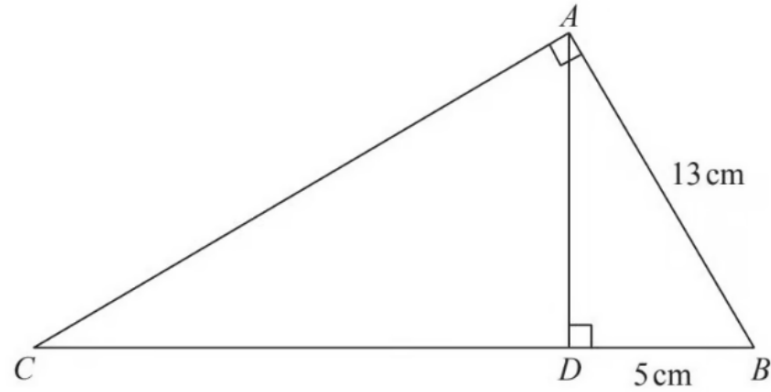
In triangle ABC , $DE \parallel BC$ with D on AB , E on AC . $AD = 4$, $DB = 6$, $DE = 5$. Find BC .

SELECTED PRACTICE

Q13 PRACTICE Practice · Easy

CH 4 · TOPIC 8 3 marks

2 ABC and ABD are two right-angled triangles.



Angle $BAC = \text{angle } ADB = 90^\circ$

$AB = 13 \text{ cm}$

$DB = 5 \text{ cm}$

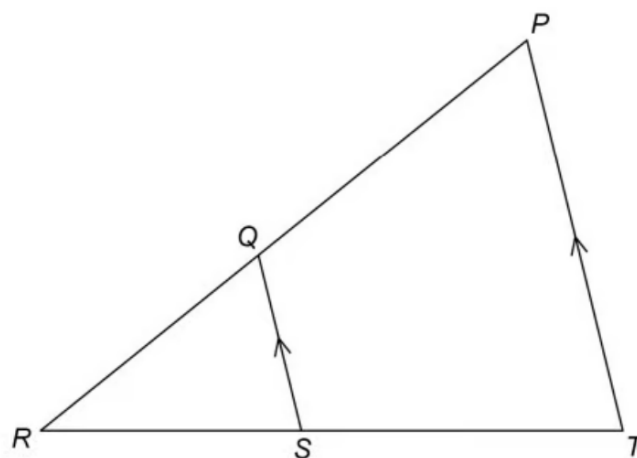
Work out the length of CB .

(3 marks)

Q14 PRACTICE Practice · Medium

CH 4 · TOPIC 8 1 mark

- 5 PRT and QRS are similar triangles.



Not drawn accurately

Which of these is equivalent to $\frac{QR}{PR}$?

Circle your answer.

$\frac{RS}{ST}$	$\frac{QS}{PT}$
$\frac{PT}{QS}$	$\frac{RT}{RS}$

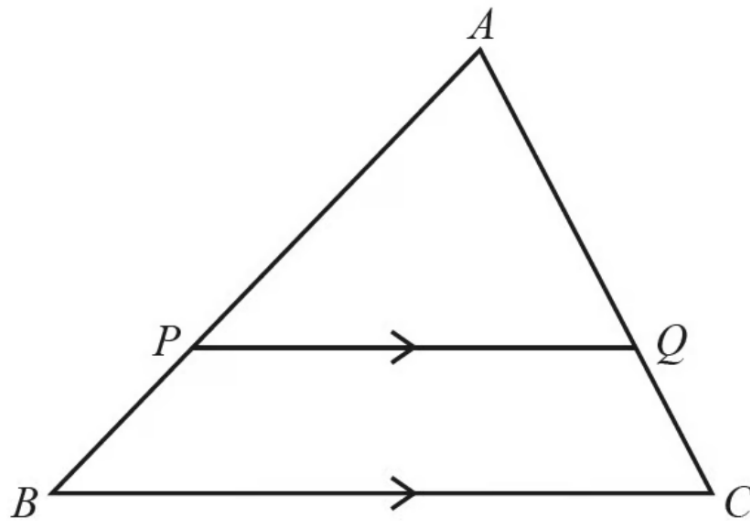
(1 mark)

Q15**PRACTICE**

Practice · Hard

CH 4 · TOPIC 8 2 marks

8

NOT TO
SCALE

In the diagram, PQ is parallel to BC .
 APB and AQC are straight lines.
 $PQ = 8$ cm, $BC = 10$ cm and $AB = 9$ cm.

Calculate PB .

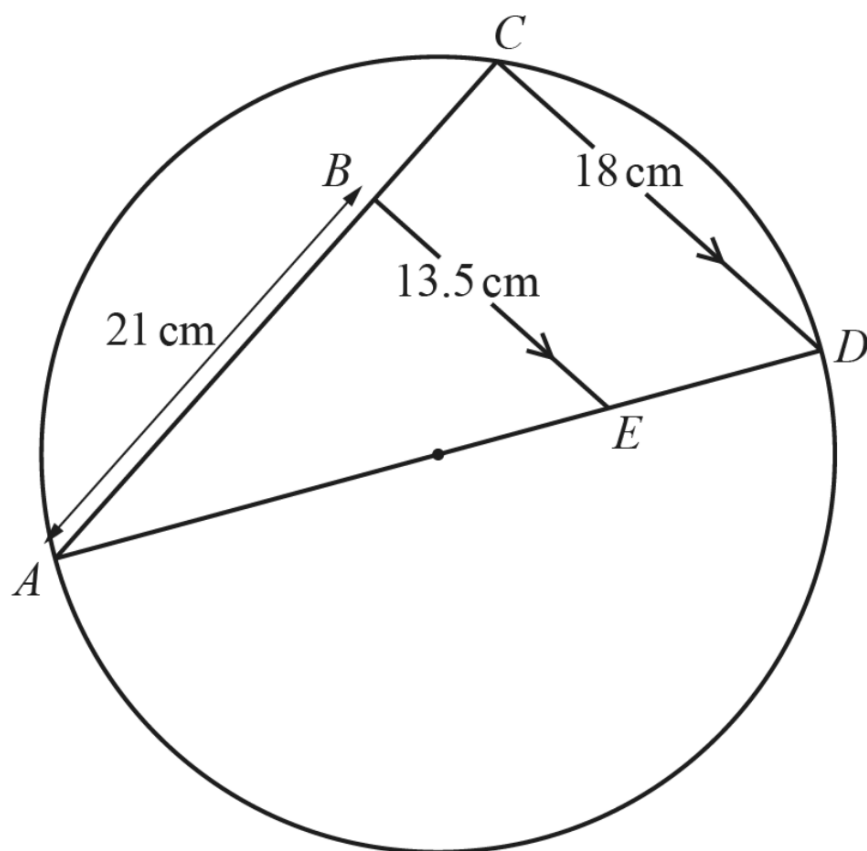
$PB = \dots\dots\dots$ cm
(2 marks)

Q16**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 8 5 marks

4

NOT TO
SCALE

C lies on a circle with diameter AD .

B lies on AC and E lies on AD such that BE is parallel to CD .

$AB = 21$ cm, $CD = 18$ cm and $BE = 13.5$ cm.

Work out the radius of the circle.

..... cm

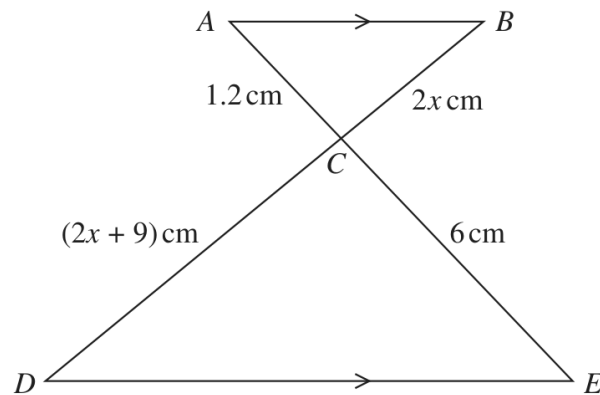
(5 marks)

PAST PAPERS

Q17 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 2HR • Q12

CH 4 • TOPIC 8 3 marks

12

Diagram **NOT**
accurately drawn

ACE and BCD are straight lines.
 AB is parallel to DE

Work out the value of x $x = \dots\dots\dots$

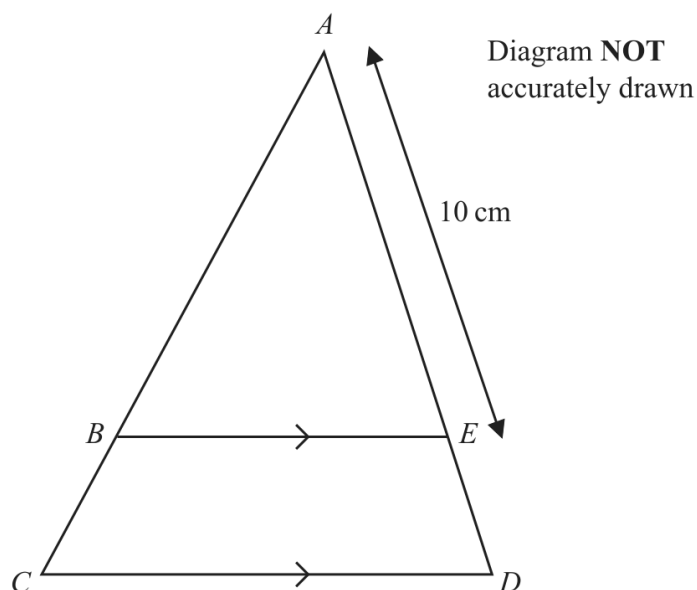
(Total for Question 12 is 3 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q12

CH 4 • TOPIC 8 4 marks

12



In the diagram, ABC and AED are straight lines.
 BE is parallel to CD

$$AE = 10 \text{ cm} \quad \text{and} \quad CD = 1.5 \times BE$$

- (a) Work out the length of ED

..... cm

$$AB = (2x + 5) \text{ cm} \text{ and } BC = (3x - 5) \text{ cm}$$

- (b) Work out the value of x

$x =$

(Total for Question 12 is 4 marks)

03 Proof: Congruence and Similarity

List three facts, each with its reason, then name the condition. Similar needs two pairs of equal angles; congruent needs SSS, SAS, ASA or RHS.

WHEN YOU SEE → YOU DO

“prove the triangles are congruent”

→ three facts + SSS / SAS / ASA / RHS

“prove the triangles are similar”

→ two pairs of equal angles, with reasons

parallel lines in the diagram

→ alternate / corresponding angles

a circle in the diagram

→ angles in the same segment

ZONE SSS

three sides

ZONE SAS

two sides +
angle between

ZONE ASA

two angles + a
side

ZONE RHS

right angle,
hypotenuse, side

WORKED MODEL • PARALLEL LINES

$AB \parallel CD$; AED and BEC are straight. Prove $ABE \sim DCE$.

1 $BAE = CDE$

alternate angles

2 $AEB = DEC$

vertically opposite angles

ANSWER two pairs of equal angles: similar

IDEA 03

Practise: Proof: Congruence and Similarity

My questions first, then selected practice, then real past papers.

Q19 DR ESLAM Congruence

CH 4 · TOPIC 8 3 marks

$ABCD$ is a parallelogram. Prove that triangles ABD and CDB are congruent.

SELECTED PRACTICE

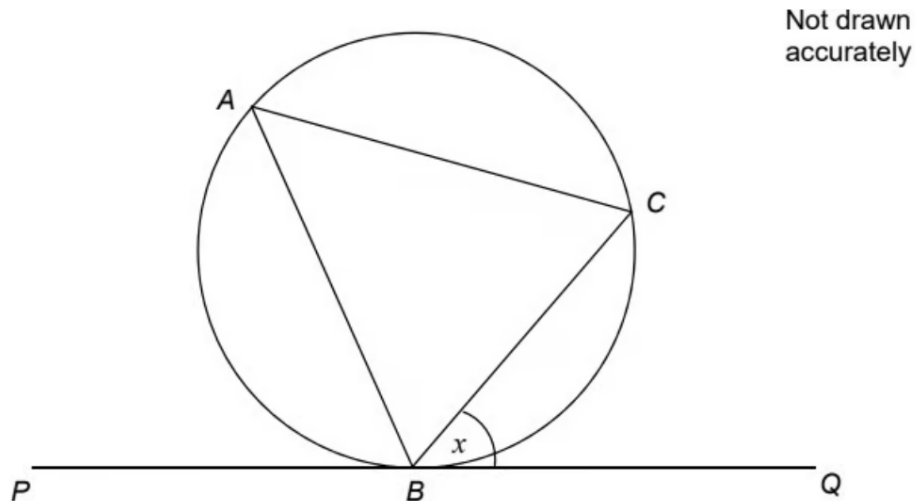
Q20**PRACTICE**

Practice · Medium

CH 4 · TOPIC 8 3 marks

6 A , B and C are points on a circle.

- BC bisects angle ABQ .
- PBQ is a tangent to the circle.



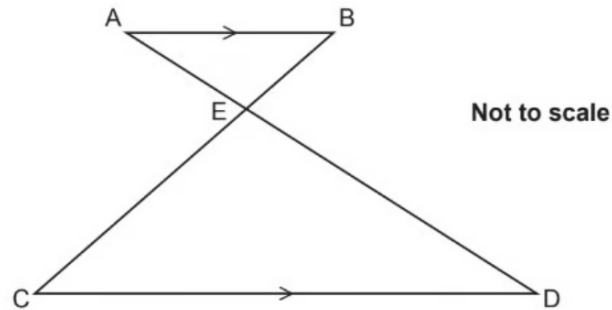
Angle $CBQ = x$

Prove that $AC = BC$

(3 marks)

Q21 **PRACTICE** Practice · MediumCH 4 · TOPIC 8 **3 marks**

- 8 In the diagram AB is parallel to CD.
AED and BEC are straight lines.



Prove that triangle ABE is similar to triangle CDE.

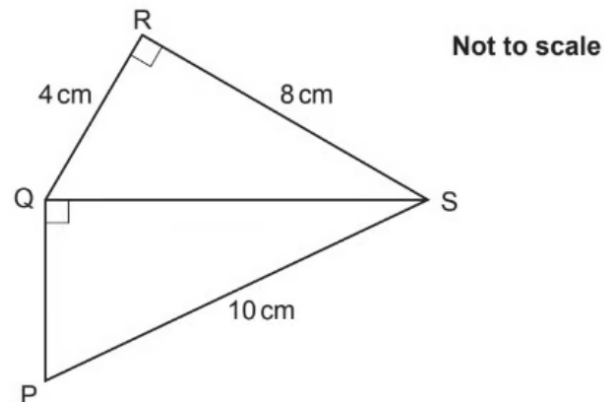
(3 marks)

Q22**PRACTICE**

Practice · Hard

CH 4 · TOPIC 8 5 marks

7 The diagram below shows two right-angled triangles.



Prove that triangles PQS and QRS are similar.

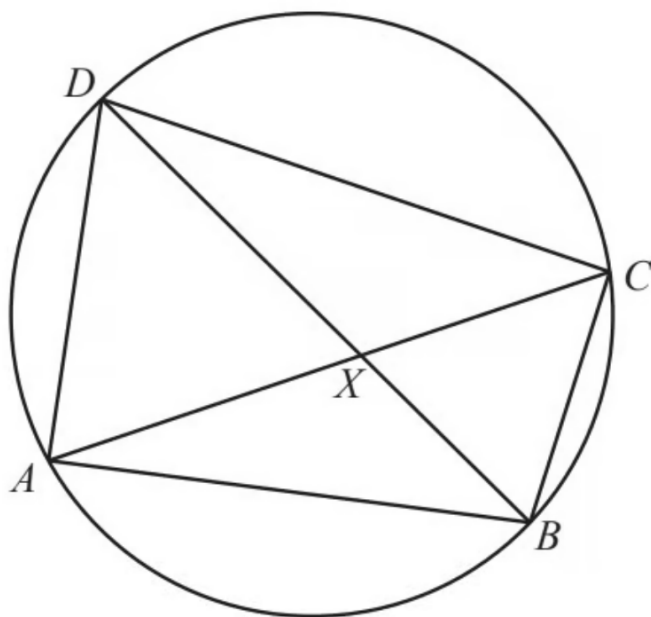
(5 marks)

Q23**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 8 5 marks

5 (a)

NOT TO
SCALE

The diagonals of the cyclic quadrilateral $ABCD$ intersect at X .

Explain why triangle ADX is similar to triangle BCX . Give a reason for each statement you make.

(3 marks)

(b) $AD = 10$ cm, $BC = 8$ cm, $BX = 5$ cm, $CX = 7$ cm.

Calculate DX .

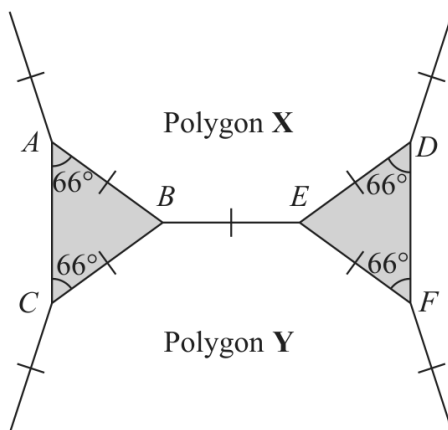
 $DX = \dots\dots\dots$ cm**(2 marks)**

PAST PAPERS

Q24 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2HR · Q12

CH 4 · TOPIC 8 3 marks

- 12 The diagram shows two congruent isosceles triangles and parts of two congruent regular polygons, **X** and **Y**.

Diagram **NOT** accurately drawn

The two regular polygons each have n sides.

Work out the value of n .

$n = \dots\dots\dots$

(Total for Question 12 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------|--|
| 01 | Scale factor | Pair by angles; $k = \text{new} \div \text{old}$. |
| 02 | In a diagram | Parallel lines $\rightarrow$ similar triangles; draw them apart. |
| 03 | Proof | Similar: two angles. Congruent: SSS / SAS / ASA / RHS. |

BOTTOM LINE Match the angles $\rightarrow$ one pair gives $k \rightarrow$ a reason for every fact in a proof.

Answers

Congruence, Similarity and Geometrical Proof • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SIMILAR SHAPES: THE SCALE FACTOR

- Q1** Dr Eslam • Scale factor $k = 2.5$: 10 cm
- Q2** Practice • Easy (a) $k = 1.5$: $LP = 7.5$ cm (b) $BC = 8$ cm
- Q3** Practice • Easy $k = \frac{16}{20}$: $a = 9.6$ cm
- Q4** Practice • Medium $147 \times \frac{90}{187} = 71$ cm
- Q5** Practice • Very Hard (a) $1200\pi = 3770$ cm³ (b) $\frac{4}{3}\pi r^3(1 + 8) = 600\pi$:
 $r = 3.68$ cm
- Q6** Edexcel 4MA1 • November 2021 • Paper 2H • Q4 (a) $AB = 81$ cm (b) $EF = 24$ cm
- Q7** Edexcel 4MA1 • January 2022 • Paper 2H • Q3 (a) $BC = 5.5$ cm (b) $y = 3x$
- Q8** Edexcel 4MA1 • May 2023 • Paper 1HR • Q3 (a) $GH = 2.5$ cm (b) $BC = 9.12$ cm
- Q9** Edexcel 4MA1 • November 2024 • Paper 1H • Q10 $DE = 45$, $k = 6$: $x = 4$
- Q10** Edexcel 4MA1 • June 2025 • Paper 2H • Q5 (a) $x = 12.5$ (b) $y = 9.6$
- Q11** Edexcel 4MA1 • November 2025 • Paper 2H • Q4 $JK = 20$ cm

IDEA 02 • SIMILAR TRIANGLES IN A DIAGRAM

- Q12** Dr Eslam • Nested triangles
 $k = \frac{10}{4}$: $BC = 12.5$
- Q13** Practice • Easy
 triangles $ABD \sim CBA$: $CB = \frac{13^2}{5} = 33.8$ cm
- Q14** Practice • Medium
 $\frac{QS}{PT}$
- Q15** Practice • Hard
 $AP = 7.2$: $PB = 1.8$ cm
- Q16** Practice • Very Hard
 $AC = 28$, $AD = \sqrt{28^2 + 18^2} = 33.3$: radius 16.6 cm
- Q17** Edexcel 4MA1 • June 2022 • Paper 2HR • Q12
 $x = 1.125$
- Q18** Edexcel 4MA1 • June 2024 • Paper 2HR • Q12
 (a) $ED = 5$ cm (b) $x = 3.75$

IDEA 03 • PROOF: CONGRUENCE AND SIMILARITY

- Q19** Dr Eslam • Congruence
 $AB = CD$, $AD = CB$ (opposite sides), BD common: SSS
- Q20** Practice • Medium
 $CBQ = x = BAC$ (alternate segment) and $ABC = x$ (bisector): base angles equal, so $AC = BC$
- Q21** Practice • Medium
 $BAE = CDE$ (alternate), $ABE = DCE$ (alternate), $AEB = DEC$ (vertically opposite): similar
- Q22** Practice • Hard
 $QS = \sqrt{4^2 + 8^2} = \sqrt{80}$; angles $PQS = QRS = 90^\circ$ and $\frac{PQ}{QR} = \frac{QS}{RS}$ with $PQ = \sqrt{100 - 80} = 2\sqrt{5}$: both $\frac{\sqrt{5}}{2}$, so similar
- Q23** Practice • Very Hard
 (a) $DAX = CBX$ and $ADX = BCX$ (same segment), $AXD = BXC$ (vertically opposite) (b) $DX = 6.25$ cm
- Q24** Edexcel 4MA1 • January 2020 • Paper 2HR • Q12
 apex angle 48° : interior angle $156^\circ \rightarrow n = 15$